

Helix OTA — Production Completion Master Index

Revision: 1 **Created:** 2026-07-26 **Status:** active — exhaustive step-by-step guide to production deployment **Authority:** §11.4.172 (living production-readiness plan) · §11.4.185 (manual QA final confirmation) · §11.4.6 (no-guessing — every claim cites a path)

Purpose

This directory contains the exhaustive, numbered, step-by-step guide to bring Helix OTA from its current state (v1.0.0 — 72/73 software gaps closed, 1 hardware-gated) to full production deployment on a remote backend host, with system images pre-loaded with the OTA client, and a complete update lifecycle proven end-to-end on real hardware.

Every step is numbered. Steps marked [OPERATOR] require human decision or action. Steps marked [AGENT] are automatable code/implementation work. Steps marked [HARDWARE] require physical RK3588/Orange Pi 5 Max board.

Reading Order

#	Document	Purpose
0	00_MASTER_INDEX.md (this file)	Master index, priority matrix, danger zones
1	01_OPERATOR_DECISIONS.md	All operator-gated decisions blocking progress
2	02_SERVER_HARDENING.md	Server production-readiness hardening
3	03_MULTI_TENANT_ACCOUNTS.md	Multi-tenant Accounts feature (XL effort)

#	Document	Purpose
4	04_WEBSITE_AND_DASHBOARD.md	Marketing website + dashboard production-readiness
5	05_DEVICE_SIDE.md	Android agent, ApplyPort, A/B, device client
6	06_DEPLOYMENT_INFRA.md	Remote deployment, monitoring, backup, TLS
7	07_SYSTEM_IMAGES.md	System image building, flashing, OTA artifact pipeline
8	08_TESTING_AND_QA.md	Full re-test, manual QA, release tagging
9	09_SUBMODULE_HYGIENE.md	Submodule manifests, mirror forks, tracking tooling

Priority Matrix (What to do and in what order)

STAGE A — OPERATOR DECISIONS (Blocking — must resolve FIRST)

These 12 operator decisions block substantial implementation work. They produce ZERO code — they produce answers. Resolve ALL before any agent-executable code work begins.

Step	Decision	Blocks	Doc
A-01	Accounts: RBAC model (hybrid-first?), identity source (local/OAuth?), id type (TEXT/UUID)	M1-M8 Accounts	01_OPERATOR_DECISIONS.md §1
A-02	Accounts: per-account signing keys, default-account backfill strategy	M1, M7	01_OPERATOR_DECISIONS.md §1
A-03	Website: repo name/remote, containerized	P4 Website	01_OPERATOR_DECISIONS.md §2

Step	Decision	Blocks	Doc
	build, tokens-only OpenDesign Mirror-fork canonicalization:		
A-04	vision_engine, llm_orchestrator, llm_provider	Submodule hygiene	01_OPERATOR_DECISIONS.md §3
A-05	DDoS default posture: HELIX_MAX_INFLIGHT default (256 vs 0)	Server security	01_OPERATOR_DECISIONS.md §4
A-06	ROLL-1: SuccessThreshold==0 canary semantics (reject or document)	Rollout engine	01_OPERATOR_DECISIONS.md §5
A-07	OTA signing metadata crypto-binding scope	Artifact security	01_OPERATOR_DECISIONS.md §6
A-08	Remote deploy host: IP/hostname, SSH auth (key vs password), product scope (Svord/ Mistiq board IDs)	Deployment	01_OPERATOR_DECISIONS.md §7
A-09	lets_encrypt + sftp submodule paths, CLI entrypoints confirmed	TLS/Artifact publishing	01_OPERATOR_DECISIONS.md §8
A-10	Website root behavior: hxota.dev serves console vs redirects to hxota.com	Website	01_OPERATOR_DECISIONS.md §9
A-11	Runtime-secret provisioning strategy (operator-provided vs auto-generated)	Deployment security	01_OPERATOR_DECISIONS.md §10
A-12	Multi-track toolkit alias provisioning (OAuth tokens for headless workers)	Parallel dev infra	01_OPERATOR_DECISIONS.md §11

STAGE B — SERVER PRODUCTION HARDENING (High priority — runs in parallel with Stage A decisions)

Step	Task	Effort	Doc
B-01	Verify SEC-1 token fail-fast is actually committed and working	Verify	02_SERVER_HARDENING.md
B-02	Fix /readyz probe to return false on DB/S3 failure	S	02_SERVER_HARDENING.md
B-03	Add one-of-pair TLS config fail-fast validation	S	02_SERVER_HARDENING.md
B-04	Add TLS to default production compose	S	02_SERVER_HARDENING.md
B-05	Add production-appropriate HELIX_MAX_INFLIGHT and rate-limit defaults	S	02_SERVER_HARDENING.md
B-06	Auth hardening: bcrypt-persist admin user seeds, durable refresh-token store	M	02_SERVER_HARDENING.md
B-07	Wire device-side min_current_version into UpdateAvailable + Kotlin DTO	M	02_SERVER_HARDENING.md
B-08	Add per-IP/token rate limiting with brute-force login guard	S	02_SERVER_HARDENING.md
B-09	Wire the security submodule into server for keyword filtering and redaction	S	02_SERVER_HARDENING.md
B-10	Verify and fix G-19 (migration Down() methods — claimed-closed but DownSQL not callable)	S	02_SERVER_HARDENING.md
B-11		S	02_SERVER_HARDENING.md

Step	Task	Effort	Doc
	HB-1 fw_setenv flush-error research-gated fix		
B-12	Wire the webhook dispatch engine to deployment lifecycle events	M	02_SERVER_HARDENING.md
B-13	Add OTA signing metadata crypto-binding (sign over OType+Board+Version)	M	02_SERVER_HARDENING.md

STAGE C — MULTI-TENANT ACCOUNTS (XL effort — the biggest single feature)

Step	Task	Effort	Doc
C-01	M1: Data model + migrations + store scoping (Account, User, Membership, account_id on all tables)	XL	03_MULTI_TENANT_ACCOUNTS.md
C-02	M2: Token account_id claim + super-admin authZ + single role vocabulary	L	03_MULTI_TENANT_ACCOUNTS.md
C-03	M3: Account-scope every OTA route + super-admin / admin/* API + GET / artifacts list	L	03_MULTI_TENANT_ACCOUNTS.md
C-04		L	03_MULTI_TENANT_ACCOUNTS.md

Step	Task	Effort	Doc
	M4: UI sign-in split + account switcher + super-admin console (dashboard + ota-manager)		
C-05	M5: Project- side integration CLI (server/ cmd/helix- ota)	M	03_MULTI_TENANT_ACCOUNTS.md
C-06	M6: Device update client + setup wizard (ota- android- agent) + per- account key verify	XL	03_MULTI_TENANT_ACCOUNTS.md
C-07	M7: Object storage seam (MinIO/S3, signed URLs, per-account signing-key registry)	L	03_MULTI_TENANT_ACCOUNTS.md
C-08	M8: Full retest + manual QA merge gate	M	03_MULTI_TENANT_ACCOUNTS.md

STAGE D — WEBSITE + DASHBOARD (L effort — marketing surface + admin UIs)

Step	Task	Effort	Doc
D-01		—	04_WEBSITE_AND_DASHBOARD.md

Step	Task	Effort	Doc
	Resolve website operator decisions (A-03, A-10)		
D-02	Create submodules/ website repo, scaffold Angular 22 SSR + Tailwind v4 + OpenDesign brand	L	04_WEBSITE_AND_DASHBOARD.md
D-03	Build content to locked spec (contact@hxota.com, no pricing, heart footer, roadmap)	M	04_WEBSITE_AND_DASHBOARD.md
D-04	§11.4.190 proofs: responsive breakpoint×engine host-render + SEO audit + light/dark	M	04_WEBSITE_AND_DASHBOARD.md
D-05	Dashboard: complete §11.4.170 host-render for all feature pages	M	04_WEBSITE_AND_DASHBOARD.md
D-06	ota-manager: complete §11.4.170 host-render for all feature pages	M	04_WEBSITE_AND_DASHBOARD.md
D-07	Deploy website + dashboards to Firebase/live host	S	04_WEBSITE_AND_DASHBOARD.md

STAGE E — DEVICE-SIDE COMPLETION (Hardware-gated)

Step	Task	Effort	Doc
E-01	[HARDWARE] Attach RK3588 / Orange Pi 5 Max board, verify ADB/ SSH reachability	—	05_DEVICE_SIDE.md
E-02	[HARDWARE] RK3588 Tier-3 on-	L	05_DEVICE_SIDE.md

Step	Task	Effort	Doc
	silicon A/B apply (vendor HAL, U-Boot, dm-verity)		
E-03	Complete Linux/U-Boot ApplyPort slot-writer + driver loop (real partitions)	M	05_DEVICE_SIDE.md
E-04	Android agent: stress/chaos/bench/memory tests, real-device instrumentation	M	05_DEVICE_SIDE.md
E-05	Android update-engine bridge: real-device tests, BootStateObserver unit tests	M	05_DEVICE_SIDE.md
E-06	Device-setup wizard UI in ota-android-agent (linked to C-06)	L	05_DEVICE_SIDE.md

STAGE F — DEPLOYMENT INFRASTRUCTURE (Remote backend ready)

Step	Task	Effort	Doc
F-01	Resolve operator deployment decisions (A-07 through A-11)	—	06_DEPLOYMENT_INFRA.md
F-02	Bootstrap remote host: hxota user, rootless podman, lingering, systemctl	M	06_DEPLOYMENT_INFRA.md

Step	Task	Effort	Doc
F-03	Incorporate lets_encrypt + sftp submodules	S	06_DEPLOYMENT_INFRA.md
F-04	Provision TLS certs for all domains (hxota.dev, hxota.com, *.hxota.dev)	M	06_DEPLOYMENT_INFRA.md
F-05	Deploy full stack (ota- server + postgres + minio + proxy) to remote host	M	06_DEPLOYMENT_INFRA.md
F-06	Configure PostgreSQL backups (pg_dump → S3, cron job)	S	06_DEPLOYMENT_INFRA.md
F-07	Wire Prometheus + Grafana + Loki monitoring stack	M	06_DEPLOYMENT_INFRA.md
F-08	Deploy dashboard, console, and website SPAs	S	06_DEPLOYMENT_INFRA.md
F-09	Set up SSH key auth, rotate credentials, lock down firewall	M	06_DEPLOYMENT_INFRA.md
F-10	Production smoke test: health	M	06_DEPLOYMENT_INFRA.md

Step	Task	Effort	Doc
	probes, artifact upload, device registration, OTA cycle		

STAGE G — SYSTEM IMAGES + OTA ARTIFACT PIPELINE

Step	Task	Effort	Doc
G-01	Build Android 15 AOSP system image for Orange Pi 5 Max (RK3588) with OTA agent pre-installed	XL	07_SYSTEM_IMAGES.md
G-02	Platform-sign the OTA agent APK, place under /system/priv-app	M	07_SYSTEM_IMAGES.md
G-03	Generate Ed25519 signing keypair, secure private key, distribute public key to server config	S	07_SYSTEM_IMAGES.md
G-04	Build first OTA artifact (ZIP_STORED, signed with Ed25519) from system image delta	M	07_SYSTEM_IMAGES.md
G-05	Upload artifact to server, create release, create deployment, initiate rollout	S	07_SYSTEM_IMAGES.md
G-06	Set up CI/CD artifact build pipeline (containerized build, auto-sign, auto-upload)	L	07_SYSTEM_IMAGES.md
G-07	Flash system image to RK3588 board, verify OTA agent runs on boot	M	07_SYSTEM_IMAGES.md
G-08	End-to-end OTA update cycle on real hardware: poll → download → verify → apply → reboot → verify new slot	L	07_SYSTEM_IMAGES.md
G-09	Test A/B rollback: corrupt next boot → auto-rollback to previous slot	M	07_SYSTEM_IMAGES.md
G-10	Set up artifact publishing pipeline (SFTP to download area, CDN if applicable)	M	07_SYSTEM_IMAGES.md

STAGE H — FULL RETEST + QA + RELEASE

Step	Task	Effort	Doc
H-01		M	08_TESTING_AND_QA.md

Step	Task	Effort	Doc
H-02	Full server test suite re-run (unit + integration + stress + chaos + fuzz + security + race) Full OTA brick test suite re-run (protocol, validator, rollout, telemetry)	M	08_TESTING_AND_QA.md
H-03	End-to-end signed artifact pipeline against live production server	M	08_TESTING_AND_QA.md
H-04	Android agent tests on real device (stress + chaos + memory + benchmark)	M	08_TESTING_AND_QA.md
H-05	Distributed DDoS resilience test against production stack	M	08_TESTING_AND_QA.md
H-06	Web-surface stress/chaos tests (dashboard + ota-manager)	M	08_TESTING_AND_QA.md
H-07	Complete §11.4.153 per-feature video recording	L	08_TESTING_AND_QA.md

Step	Task	Effort	Doc
	evidence (durable committed MP4s)		
H-08	Independent code review of all new/ changed code → zero-finding GO	M	08_TESTING_AND_QA.md
H-09	[OPERATOR] §11.4.185 Manual QA- team final confirmation on physical RK3588 board	L	08_TESTING_AND_QA.md
H-10	Tag release (helix_ota-1.1.0 or 2.0.0) on main + all touched submodules	S	08_TESTING_AND_QA.md

STAGE I — SUBMODULE + TRACKING HYGIENE (Ongoing)

Step	Task	Effort	Doc
I-01	Add helix- deps.yaml to all 6 ota-* bricks	S	09_SUBMODULE_HYGIENE.md
I-02	Add upstreams/ recipes to all 6 ota- * bricks	S	09_SUBMODULE_HYGIENE.md
I-03	Add wire schema- version field to ota-protocol + ota- telemetry-schema	M	09_SUBMODULE_HYGIENE.md
I-04	Resolve mirror- fork canonicalization	M	09_SUBMODULE_HYGIENE.md

Step	Task	Effort	Doc
	(A-04), UNION-merge divergent branches		
I-05	Adopt constitution workable-items engine (replace local simplified copy)	M	09_SUBMODULE_HYGIENE.md
I-06	Extend workable_items.db schema (created_by, assigned_to, logic_groups, test_diary, reopens_count)	M	09_SUBMODULE_HYGIENE.md
I-07	Fix §11.4.33 type-aware closure vocabulary (12 items currently wrong)	S	09_SUBMODULE_HYGIENE.md
I-08	Reconcile track/branch drift, record canonical bindings in registry	S	09_SUBMODULE_HYGIENE.md
I-09	Bump llm_orchestrator parent gitlink after mirror-fork resolution	S	09_SUBMODULE_HYGIENE.md
I-10	Fix stale comments (K5 VAL-1, K6 fabric-lease)	S	09_SUBMODULE_HYGIENE.md
I-11	Remove LLMPProvider symlink after verifying no path dependency	S	09_SUBMODULE_HYGIENE.md

Cumulative Effort Estimate

Stage	Effort (T-shirt)	Operator-Dependent Steps
A — Operator Decisions	0 (decision only)	12
B — Server Hardening	M–L	0
C — Multi- Tenant Accounts	XL	1 (K12 decisions required first)
D — Website + Dashboard	L	2 (A-03, A-10)
E — Device- Side	L–XL	1 (hardware attachment)
F — Deployment Infra	M–L	5 (A-07 through A-11)
G — System Images + OTA Pipeline	XL	0 (all agent-excutable after hardware)
H — Full Retest + QA	L	1 (§11.4.185 manual QA)
I — Submodule Hygiene	M	0
TOTAL	~3XL + 3L + 3M	22 operator decisions/actions

Danger Zones (must NOT skip or shortcut)

#	Danger	Why critical
DZ-1	Deploying without Accounts → single- tenant only, no customer separation	All data shared; total rework when Accounts lands
DZ-2		

#	Danger	Why critical
	Deploying without real object storage → artifacts discarded after upload	The OTA loop is provably broken: no device can download
DZ-3	Deploying with default token secret → total token forgery	Attacker mints admin tokens at will
DZ-4	Skipping physical RK3588 validation → A/B slot switch unproven on real silicon	The emulator is not the silicon; dm-verity, vendor HAL, U-Boot differ
DZ-5	No backup strategy → data loss on PostgreSQL failure	Fleet state, device registry, artifact metadata — unrecoverable
DZ-6	No monitoring/alerting → blind to production failures	Fleet-wide OTA at scale without observability = operational catastrophe
DZ-7	Force-push to any repo/submodule → history loss	Constitution §11.4.113 strictly forbids; irreversible across 4 mirrors
DZ-8	Deploying from feature branch without merge to main	§11.4.167/§11.4.188: feature → main merge ONLY after operator approval + full retest
DZ-9	VEL-1 stall (velocity < 3/week for 2+ weeks) → operator resource escalation	73+ workable items; if closure stalls, the project stalls

What the 1.0.0 Gap Tracker Misses (genuine unclosed items)

The `gap_tracker.csv` correctly shows 72/73 entries as “Closed” but these are REMEDIATION STATUS closures (each gap had a fix or documentation applied). The following items need SUBSTANTIAL NEW WORK before production, not captured as gaps:

1. **Multi-tenant Accounts** — Entire M1–M8 delivery plan. The gap tracker treats design-completion as “Closed” but zero production code exists for the tenant layer.
 2. **Marketing Website** — Gap tracker has no entry for this (§11.4.190 mandate). Zero code exists.
 3. **Physical RK3588 validation** — G-11 is the only remaining “Queued” gap, capturing the intentional design decision (artifact download not on control plane). OTA-004 (Tier-3 hardware) is captured in Issues.md.
 4. **System image build pipeline** — No gap tracker entry for producing the actual Android system images with pre-installed OTA client.
 5. **End-to-end OTA on real hardware** — Emulated-only proven; real-silicon unproven.
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How to Use This Guide

1. **Start with 01_OPERATOR_DECISIONS.md** — read it, answer every question. Until A-01 through A-12 are resolved, Stages C, D, E, and F are gated.
2. **Run Stage B in parallel** with operator decisions — these are small, independent server fixes.
3. **When A-01..A-02 are resolved, start Stage C (Accounts)** — this is the critical path.
4. **When A-03/A-10 are resolved, start Stage D (Website)** — independent work-stream.
5. **When hardware arrives, start Stage E (Device-Side)** — independent of Accounts/Website.
6. **When A-07..A-11 are resolved, start Stage F (Deployment)** — can run alongside C/D.
7. **Stage G starts after C-07 (object storage) + E-01 (hardware) are done.**
8. **Stage H is the final gate** — nothing deploys to production until H-09 (§11.4.185 manual QA) passes.

9. **Stage I is ongoing hygiene** — perform throughout, don't batch at the end.

Honest Boundary (§11.4.6)

This guide is a synthesis of: - docs/planning/
PRODUCTION_READINESS_PLAN.md (Rev 1, 2026-07-12) - docs/planning/
DELTA_ANALYSIS_20260712.md (Rev 1, 2026-07-12) - docs/research/
production_planning_20260726/ANALYSIS.md + gap_tracker.csv - docs/
research/accounts/30_delivery_plan.md (Rev 1, 2026-07-10) - docs/
remote_deploy/REMOTE_DEPLOY.md (Rev 1, 2026-07-14) - docs/
RESUMPTION.md (Rev 12, 2026-07-26) - Static-read verification of server
code, route wiring, migration framework - Submodule state (20
submodules, some with mirror-fork divergence)

Every step ID references a source document. Effort estimates are T-shirt sizes (S/M/L/XL), not calendar dates. §11.4.172 forbids false precision until velocity is measured over ≥ 3 data points.